

Literature Status: Mackey Completeness Does Not Force Lie-Group Regularity

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Status: literature already answered, negatively. Problem II.2 of Neeb asks whether every Lie group modelled on a Mackey-complete locally convex space is regular. Han and Liu construct a complex analytic BCH–Lie group modelled on a complete Silva space that is not even C^0 -semiregular. Completeness implies Mackey completeness, so their example is a counterexample to the stated problem.

1 Original question

Karl-Hermann Neeb, *Towards a Lie theory of locally convex groups*, arXiv:1501.06269 [1], asks on arXiv PDF page 40:

Problem II.2. Show that every Lie group G modeled on a Mackey complete locally convex space is regular, or find a counterexample.

Regularity requires, in particular, existence of the group-valued evolution for every smooth Lie-algebra curve.

2 Separate later counterexample

Zongjian Han and Fungo Liu, *Completeness of the Model Space Does Not Force Regularity for Infinite-Dimensional Lie Groups*, arXiv:2607.28557 [2], prove in Theorem 1.1 (arXiv PDF page 2) that there is a contractible complex analytic BCH–Lie group H , modelled on the complete Silva space $\varphi_{\mathbb{C}}$, whose exponential map is a homeomorphism but which is not C^0 -semiregular. More concretely, a fixed finite-dimensional subspace $W \subseteq L(H)$ contains smooth controls

$$\gamma_\lambda \in C^\infty([0, 1], W)$$

for which the evolution equation has no H -valued C^1 solution. Their Theorem 1.2 gives the construction as the principal unit group of the unitization of a positively graded infinite-dimensional nil algebra; the failure is caused by products escaping to arbitrarily high homogeneous degrees.

This matches the source hypotheses exactly. A complete locally convex space is Mackey complete, and a failure of evolution existence for even one smooth control rules out Milnor regularity. Hence H is the counterexample demanded by Problem II.2. In fact, the example is stronger than requested: the model space is complete (not merely Mackey complete), and H is BCH with a globally well-behaved exponential map.

3 Scope and verification

This is a direct literature identification, not a new theorem of this run. The source question was checked in the original PDF, and the later theorem and its construction were checked in the separate supporting manuscript. The result answers Problem II.2 only. As the answering paper itself notes, the weaker Problem II.1—whether Mackey completeness forces the existence of a smooth exponential map—remains open, because its counterexample does have an exponential map.

References

- [1] K.-H. Neeb, *Towards a Lie theory of locally convex groups*, arXiv:1501.06269 (2015).
- [2] Z. Han and F. Liu, *Completeness of the Model Space Does Not Force Regularity for Infinite-Dimensional Lie Groups*, arXiv:2607.28557 (2026).